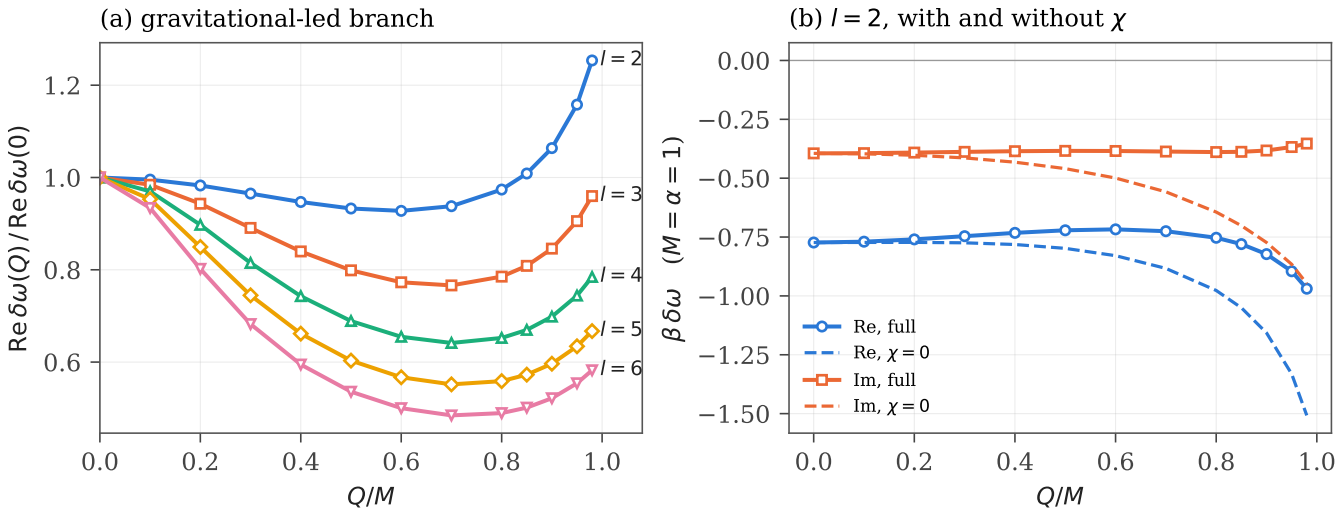
(a) gravitational-led branch
$\mathrm{Re}\,\delta\omega(Q)\,/\,\mathrm{Re}\,\delta\omega(0)$
1.2
1.0
0.8
0.6
0.0
0.2
0.4
0.6
0.8
1.0
Q/M
$l=2$
$l=3$
$l=4$
$l=5$
$l=6$
(b) $l=2$, with and without χ
$\beta\,\delta\omega \quad (M=\alpha=1)$
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
0.0
0.2
0.4
0.6
0.8
1.0
Q/M
Re, full
Re, $\chi=0$
Im, full
Im, $\chi=0$